i.MX8DXL and i.MX95 SHE(Secure Hardware Extension) API Rev 0.1

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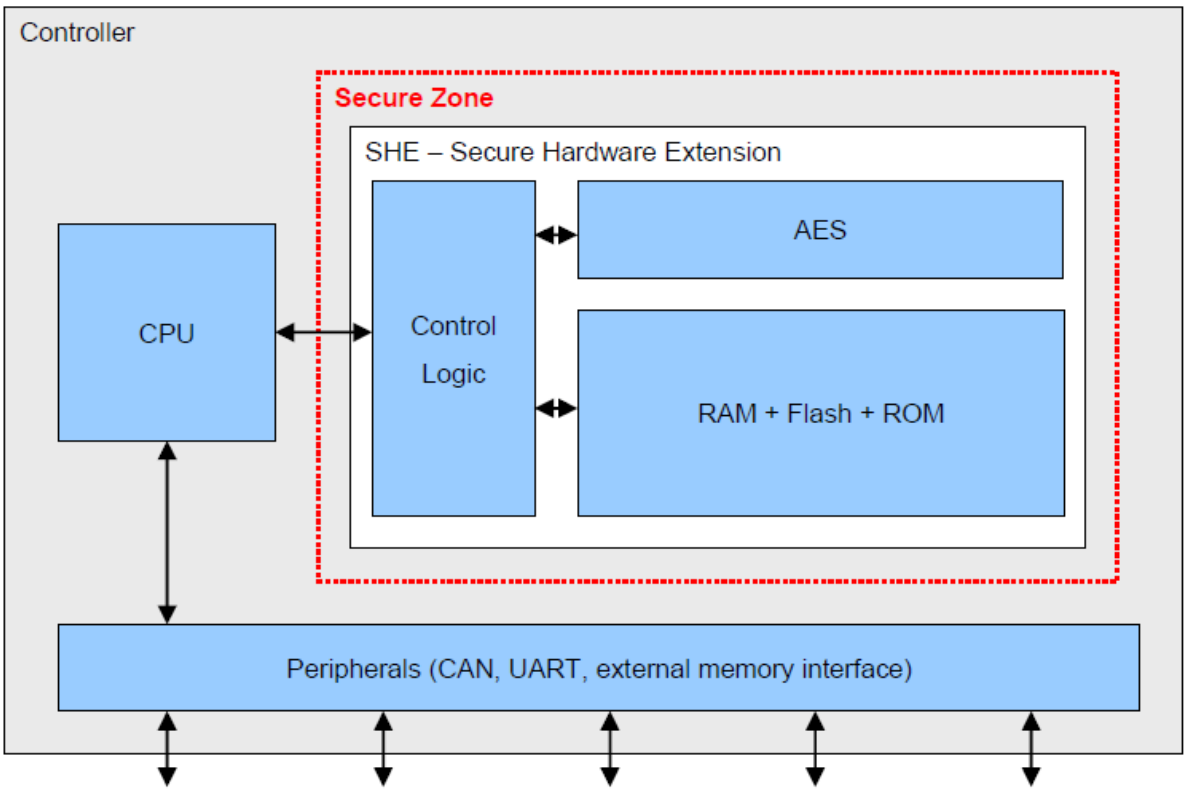
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Chapter 1

SHE(Secure Hardware Extension) API

This document is a software reference description of the API provided by the Secure Enclave Library on i.MX8DXL and i.MX95 platforms for SHE solutions



Chapter 2

Revision History

Revision	date	description
0.1	Jan 13 2024	Initial Draft
0.1	Apr 1 2024	Remove key_ext variable from key_update structure
0.2	Apr 15 2024	Update Global Info for i.MX95 Lifecycle values
0.3	Aug 28 2024	Update Global Info and get_soc_info ioctl for board_type
0.4	Nov 14 2024	Add support of CMAC API

Chapter 3

General concepts related to the API

3.1 Session

The API must be initialized by a potential requestor by opening a session. The session establishes a route (MU, DomainID...) between the requester and the SHE. When a session is opened, the SHE returns a handle identifying the session to the requester.

3.2 Service flow

For a given category of services which require service handle, the requestor is expected to open a service flow by invoking the appropriate SHE API. The session handle, as well as the control data needed for the service flow, are provided as parameters of the call. Upon reception of the open request, the SHE allocates a context in which the session handle,as well as the provided control parameters are stored and return a handle identifying the service flow.
The context is preserved until the service flow, or the session, are closed by the user and it is used by the SHE to proceed with the sub-sequent operations requested by the user on the service flow.

3.3 Key store

A key store can be created by specifying the CREATE flag in the she_open_key_store_service API. Please note that the created key store will be not stored in the NVM till a key is generated or imported specyfing the "STRICT OPERATION" flag.

3.4 Implementation specificities

SHE API with common features are supported on i.MX8DXL and i.MX95.The details of supported features per chip will be listed in the platform specifities.

- for i.MX8DXL: IMX8DXL
- for i.MX95: IMX95

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4.1 Modules

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Data Structure Index

5.1 Data Structures

Here are the data structures with brief descriptions:

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Chapter 6

Module Documentation

6.1 CMD_ENC_CBC / CMD_DEC_CBC and CMD_ENC_ECB / CMD_DEC_ECB

Data Structures

- struct open_svc_cipher_args_t
- struct op_cipher_args_t

Macros

- #define SHE_CIPHER_ONE_GO_ALGO_AES_ECB ((she_op_cipher_one_go_algo_t)(0x00u))

  Indicates it is AES ECB Cipher operation.
- #define SHE_CIPHER_ONE_GO_ALGO_AES_CBC ((she_op_cipher_one_go_algo_t)(0x01u))

  Indicates it is AES CBC Cipher operation.
- #define SHE_CIPHER_ONE_GO_FLAGS_DECRYPT ((she_op_cipher_one_go_flags_t)(0u << 0))

  Bit indicating the decrypt operation.
- #define SHE_CIPHER_ONE_GO_FLAGS_ENCRYPT ((she_op_cipher_one_go_flags_t)(1u << 0))

  Bit indicating the encrypt operation.

Typedefs

- typedef uint8_t she_op_cipher_one_go_algo_t

  Bit field indicating the requested cipher operations.
- typedef uint8_t she_op_cipher_one_go_flags_t

  Bit field indicating the requested encrypt/decrypt operations.
- typedef op_cipher_args_t op_cipher_one_go_args_t

Functions

- she_err_t she_open_cipher_service (she_hdl_t session_hdl, open_svc_cipher_args_t *args)
- she_err_t she_close_cipher_service (she_hdl_t cipher_handle)
- she_err_t she_cipher_one_go (she_hdl_t cipher_handle, op_cipher_one_go_args_t *args)

6.1.1 Detailed Description

SHE supports two modes electronic cipher book mode (ECB) for processing single blocks of data and the cipher block chaining mode (CBC) for processing larger amounts of data.

6.1.2 Data Structure Documentation

6.1.2.1 struct open_svc_cipher_args_t

Structure describing the open cipher service members

Data Fields

uint32_t	cipher_hdl	handle identifying the cipher service flow
uint8_t	flags	bitmap specifying the services properties
uint8_t	reserved[3]	reserved bits

6.1.2.2 struct op_cipher_args_t

Structure describing the cipher one go operation arguments

Data Fields

uint32_t	key_identifier	identifier of the key to be used for the operation
uint8_t *	iv	pointer to the initialization vector (nonce in case of AES CCM)
uint16_t	iv_size	It must be 12 for AES in CCM mode (not valid for SHE) length in bytes of the initialization vector. it must be 0 for algorithms not using the initialization vector.
uint8_t	svc_flags	bitmap specifying the services properties.
uint8_t	flags	bitmap specifying the operation attributes
uint8_t	cipher_algo	algorithm to be used for the operation
uint8_t *	input	Note: In case of CCM it is the purported ciphertext. pointer to the input area: • plaintext for encryption • ciphertext for decryption
uint8_t *	output	pointer to the output area: • ciphertext for encryption Note: In case of CCM it is the output of the generation-encryption process. • plaintext for decryption
uint32_t	input_size	length in bytes of the input. • In case of CBC and ECB, the input size should be multiple of a block cipher size (16 bytes).
uint32_t	output_size	length in bytes of the output

6.1.3 Typedef Documentation

6.1.3.1 op_cipher_one_go_args_t

```
typedef op_cipher_args_t op_cipher_one_go_args_t
```

Structure describing the cipher operation arguments

6.1.4 Function Documentation

6.1.4.1 she_open_cipher_service()

```
she_err_t she_open_cipher_service (
            she_hdl_t session_hdl,
            open_svc_cipher_args_t * args )
```

Open a cipher service flow. User can call this function only after having opened a key-store service flow. User must open this service in order to perform cipher operation.

Parameters

session_hdl	handle identifying the SHE session.
args	pointer to the structure containing the function arguments.

Returns

error code.

6.1.4.2 she_close_cipher_service()

```
she_err_t she_close_cipher_service (
            she_hdl_t cipher_handle )
```

Terminate a previously opened cipher service flow

Parameters

cipher_handle	handle identifying the Cipher service.

Returns

error code.

6.1.4.3 she_cipher_one_go()

```
she_err_t she_cipher_one_go (
            she_hdl_t cipher_handle,
            op_cipher_one_go_args_t * args )
```

Perform ciphering operation i.e.

CBC encryption/decryption and ECB encryption/decryption of a given plaintext/ciphertext with the key identified by key_id.

User can call this function only after having opened a cipher service flow

Parameters

cipher_handle	handle identifying the cipher service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.2 CMD_EXPORT_RAM_KEY

Data Structures

- struct op_export_plain_key_args_t

Functions

- she_err_t she_export_plain_key (she_hdl_t utils_handle, op_export_plain_key_args_t *args)

6.2.1 Detailed Description

The function exports the RAM_KEY into a format protected by SECRET_KEY. The key can be imported again by using CMD_LOAD_KEY. A RAM_KEY can only be exported if it was written into SHE in plaintext

6.2.2 Data Structure Documentation

6.2.2.1 struct op_export_plain_key_args_t

Structure describing the export RAM key operation arguments

Data Fields

uint8_t *	m1	pointer to the output address for M1 message
uint8_t	m1_size	size of M1 message - 128 bits
uint8_t *	m2	pointer to the output address for M2 message
uint8_t	m2_size	size of M2 message - 256 bits
uint8_t *	m3	pointer to the output address for M3 message
uint8_t	m3_size	size of M3 message - 128 bits
uint8_t *	m4	pointer to the output address for M4 message
uint8_t	m4_size	size of M4 message - 256 bits
uint8_t *	m5	pointer to the output address for M5 message
uint8_t	m5_size	size of M5 message - 128 bits

6.2.3 Function Documentation

6.2.3.1 she_export_plain_key()

```
she_err_t she_export_plain_key (
            she_hdl_t utils_handle,
            op_export_plain_key_args_t * args )
```

exports the RAM_KEY into a format protected by SECRET_KEY.

Parameters

utils_handle	handle identifying the SHE utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.3 CMD_GENERATE_MAC

Data Structures

- struct op_generate_mac_t

Macros

- #define SHE_MAC_SIZE 16u

  size of the MAC generated is 128bits.
- #define SHE_MESSAGE_OFFSET 16u

  message is always kept at offset 16 for FAST MAC V2 API.
- #define SHE_FAST_MAC_FLAGS_GENERATION 0

  Flag to generate MAC.

Functions

- she_err_t she_generate_mac (she_hdl_t utils_handle, op_generate_mac_t *args)
- she_err_t she_generate_fast_mac_mubuff_v2 (she_hdl_t utils_handle, op_generate_mac_t *args)

6.3.1 Detailed Description

SHE supports two types of functions to generate MAC. 1: for 16 bytes message 2: V2-for 16 or greater than 16 bytes message.

6.3.2 Data Structure Documentation

6.3.2.1 struct op_generate_mac_t

Structure describing the FAST MAC generation operation arguments

Data Fields

uint8_t	flags	optional flag to identify the operation(generate/verify)
uint16_t	key_ext	identifier of the key extension to be used for the operation
uint16_t	key_id	identifier of the key to be used for the operation
uint16_t	message_length	length in bytes of the input message. The message is padded to be a multiple of 128 bits by SHE
uint8_t *	message	pointer to the message to be processed
uint8_t *	mac	pointer to where the output MAC should be written (128bits should be allocated there)

6.3.3 Function Documentation

6.3.3.1 she_generate_mac()

```
she_err_t she_generate_mac (
            she_hdl_t utils_handle,
            op_generate_mac_t * args )
```

Generates a MAC of a given message with the help of a key identified by key_id.

Parameters

utils_handle	handle identifying the utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.3.3.2 she_generate_fast_mac_mubuff_v2()

```
she_err_t she_generate_fast_mac_mubuff_v2 (
            she_hdl_t utils_handle,
            op_generate_mac_t * args )
```

Generates a MAC (V2) of a given message (supports length greater than 16 bytes) with the help of a key identified by key_id.

The MAC and the message are transferred to or from V2X via the SHE Messaging Unit buffer at the following system address:

- For i.MX95: SHE0: 256 bytes SHE1: 64 bytes SHE MU buffer must be: MAC(16 bytes) || Message

Verify: concatenation of expected MAC and message

Parameters

utils_handle	handle identifying the utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.4 CMD_VERIFY_MAC

Data Structures

- struct op_verify_mac_t

Macros

- #define SHE_FAST_MAC_FLAGS_VERIFICATION 1

  Flag to verify MAC.
- #define SHE_FAST_MAC_FLAGS_VERIF_BIT_LEN 2

  Flag to verify variable length MAC.
- #define SHE_FAST_MAC_VERIFICATION_STATUS_OK 0x5a3cc3a5

  verification status of MAC
- #define MAC_BYTES_LENGTH 0

  MAC length expressed in bytes.
- #define MAC_BITS_LENGTH 1

  MAC length expressed in bits.
- #define SHE_MAC_VERIFICATION_SUCCESS 0

  indication of mac verification success
- #define SHE_MAC_VERIFICATION_FAILED 1

  indication of mac verification failure

Functions

- she_err_t she_verify_mac (she_hdl_t utils_handle, op_verify_mac_t *args)
- she_err_t she_verify_fast_mac_mubuff_v2 (she_hdl_t utils_handle, op_verify_mac_t *args)

6.4.1 Detailed Description

SHE supports two types of functions to verify MAC. 1: for 16 bytes message 2: V2-for 16 or greater than 16 bytes message.

6.4.2 Data Structure Documentation

6.4.2.1 struct op_verify_mac_t

Structure describing the FAST MAC generation operation arguments

Data Fields

uint8_t	flags	optional flag to identify the operation(generate/verify)
uint16_t	key_ext	identifier of the key extension to be used for the operation
uint16_t	key_id	identifier of the key to be used for the operation
uint16_t	message_length	length in bytes of the input message. The message is padded to be a multiple of 128 bits by SHE
uint8_t *	message	pointer to the message to be processed
uint8_t *	mac	pointer to the MAC to be compared
uint8_t	mac_length	number of MAC bytes to be compared with the expected value. It cannot be lower than 4 bytes.
uint32_t	verification_status	result of the MAC comparison
uint8_t	mac_length_encoding	mac length expressed in bytes or bits

6.4.3 Function Documentation

6.4.3.1 she_verify_mac()

```
she_err_t she_verify_mac (
            she_hdl_t utils_handle,
            op_verify_mac_t * args )
```

Verify the MAC of a given message with the help of a key identified by key_id.

Parameters

utils_handle	handle identifying the utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.4.3.2 she_verify_fast_mac_mubuff_v2()

```
she_err_t she_verify_fast_mac_mubuff_v2 (
            she_hdl_t utils_handle,
            op_verify_mac_t * args )
```

Verify the MAC (V2) of a given message (supports message length greater than 16 bytes) with the help of a key identified by key_id.

Parameters

utils_handle	handle identifying the utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.5 CMD_GET_ID

Data Structures

- struct op_get_id_args_t

Macros

- #define SHE_CHALLENGE_SIZE 16u

  size of the input challenge vector is 128 bits.
- #define SHE_ID_SIZE 15u

  size of the Identity(ID) returned is 120 bits.
- #define SHE_MAC_SIZE 16u

  size of the computed MAC is 128 bits.

Functions

- she_err_t she_get_id (she_hdl_t utils_handle, op_get_id_args_t *args)

6.5.1 Detailed Description

Status register can be read using below command, its an 8 bit register and has very useful information (more bit description, please refer SHE specification)

6.5.2 Data Structure Documentation

6.5.2.1 struct op_get_id_args_t

Structure describing the GET ID operation arguments

Data Fields

uint8_t	challenge[SHE_CHALLENGE_SIZE]	input Challenge vector
uint8_t	id[SHE_ID_SIZE]	identity (UID) returned by the command
uint8_t	sreg	status register returned by the command
uint8_t	mac[SHE_MAC_SIZE]	MAC returned by the command.

6.5.3 Function Documentation

6.5.3.1 she_get_id()

```
she_err_t she_get_id (
            she_hdl_t utils_handle,
```

op_get_id_args_t * *args*)

This function returns the identity (UID) and the value of the status register protected by a MAC over a challenge and the data. User can call this function only after opening the utility service.

Parameters

utils_handle	handle identifying the utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.6 SHE Get Info

Get miscellaneous information.

Data Structures

- struct op_get_info_args_t

Functions

- she_err_t she_get_info (she_hdl_t session_hdl, op_get_info_args_t *args)

6.6.1 Detailed Description

Get miscellaneous information.

This function return, among others, all the information needed to build a valid signed message.

6.6.2 Data Structure Documentation

6.6.2.1 struct op_get_info_args_t

Structure describing the get info operation member arguments

Data Fields

uint32_t	user_sab_id	Stores User identifier (32bits)
uint8_t *	chip_unique_id	Stores the chip unique identifier.
uint16_t	chip_unq_id_sz	Size of the chip unique identifier in bytes.
uint16_t	chip_monotonic_counter	Stores the chip monotonic counter value (16bits)
uint16_t	chip_life_cycle	Stores the chip current life cycle bitfield (16bits)
uint32_t	version	Stores the module version (32bits)
uint32_t	version_ext	Stores the module extended version (32bits)
uint8_t	fips_mode	Stores the FIPS mode bitfield (8bits). Bitmask definition: bit0 - FIPS mode of operation: • value 0 - part is running in FIPS non-approved mode. • value 1 - part is running in FIPS approved mode. bit1 - FIPS certified part: • value 0 - part is not FIPS certified. • value 1 - part is FIPS certified. bit2-7: reserved • value 0.

6.6.3 Function Documentation

6.6.3.1 she_get_info()

```
she_err_t she_get_info (
            she_hdl_t session_hdl,
            op_get_info_args_t * args )
```

User can call this function only after having opened the SHE session.

Parameters

session_hdl	handle identifying the active session.
args	pointer to the structure containing the function arguments.

Returns

error code

6.7 SHE Commands

Modules

- CMD_ENC_CBC / CMD_DEC_CBC and CMD_ENC_ECB / CMD_DEC_ECB
- CMD_EXPORT_RAM_KEY
- CMD_GENERATE_MAC
- CMD_VERIFY_MAC
- CMD_GET_ID
- CMD_GET_STATUS
- CMD_LOAD_KEY
- CMD_LOAD_PLAIN_KEY
- CMD_GENERATE_CMAC and CMD_VERIFY_CMAC
- CMD_INIT_RNG
- CMD_RND
- CMD_EXTEND_SEED
- last rating code
- CMD_CANCEL

6.7.1 Detailed Description

6.8 CMD_GET_STATUS

Data Structures

- struct op_get_status_args_t

Functions

- she_err_t she_get_status (she_hdl_t utils_handle, op_get_status_args_t *args)

6.8.1 Detailed Description

Return the content of status register

6.8.2 Data Structure Documentation

6.8.2.1 struct op_get_status_args_t

Structure describing the get status operation arguments

Data Fields

uint8_t	sreg	status register bits
uint8_t	pad[3]	padding bytes

6.8.3 Function Documentation

6.8.3.1 she_get_status()

```
she_err_t she_get_status (
            she_hdl_t utils_handle,
            op_get_status_args_t * args )
```

Command to get the content of the status register

Parameters

utils_handle	handle identifying the utils service
args	pointer to the structure containing the function arguments.

Returns

error code.

6.9 Session

SHE session is the first session to be opened, all other sessions are based on it.

Data Structures

- struct she_session_hdl_s
- struct she_service_hdl_s
- struct open_session_args_t

Macros

- #define SHE_HANDLE_NONE (0x0)

  Handle not available.
- #define SHE_MAX_SESSIONS (8u)

  Maximum sessions supported.
- #define SHE_MAX_SERVICES (32u)

  Maximum services supported.
- #define MAX_KEY_STORE_SESSIONS (5u)

  Maximum Key store sessions supported.
- #define SHE_OPEN_SESSION_PRIORITY_LOW (0x00U)

  Low priority session, default setting on platforms that doesn't suppor sessions priorities.
- #define SHE_OPEN_SESSION_PRIORITY_HIGH (0x01U)

  High Priority session.
- #define SHE_OPEN_SESSION_FIPS_MODE_MASK BIT(0)

  Only FIPS certified operations authorized in this session.
- #define SHE_OPEN_SESSION_EXCLUSIVE_MASK BIT(1)

  No other SHE session will be authorized on the same security enclave.
- #define SHE_OPEN_SESSION_LOW_LATENCY_MASK BIT(3)

  Use a low latency SHE implementation.
- #define SHE_OPEN_SESSION_NO_KEY_STORE_MASK BIT(4)

  No key store will be attached to this session. May provide better performances on some operation depending on the implementation. Usage of the session will be restricted to operations that doesn't involve secret keys (e.g. random generation)

Typedefs

- typedef uint32_t she_hdl_t

Functions

- struct she_session_hdl_s * she_session_hdl_to_ptr (uint32_t hdl)
- void delete_she_session (struct she_session_hdl_s *s_ptr)
- struct she_session_hdl_s * add_she_session (void)
- struct she_service_hdl_s * she_service_hdl_to_ptr (uint32_t hdl)
- void delete_she_service (struct she_service_hdl_s *s_ptr)
- struct she_service_hdl_s * add_she_service (struct she_session_hdl_s *session)
- she_err_t she_open_session (open_session_args_t *args, she_hdl_t *session_hdl)
- she_err_t she_close_session (she_hdl_t session_hdl)

6.9.1 Detailed Description

SHE session is the first session to be opened, all other sessions are based on it.

All functions and related data types required for opening a SHE session are described below.

6.9.2 Data Structure Documentation

6.9.2.1 struct she_session_hdl_s

Structure describing the session handle members

Data Fields

struct plat_os_abs_hdl *	phdl	Pointer to OS device node.
uint32_t	session_hdl	Session handle.
uint32_t	mu_type	Session MU type.
uint32_t	last_rating	last error code returned by command.

6.9.2.2 struct she_service_hdl_s

Structure describing the service handle members

Data Fields

struct she_session_hdl_s *	session	Pointer to session handle.
uint32_t	service_hdl	Service handle.

6.9.2.3 struct open_session_args_t

Structure detailing the open session operation member arguments

Data Fields

uint32_t	session_hdl	Session handle.
uint8_t	mu_type	MU with associated functionality on which session will be opened.
uint8_t	interrupt_idx	Interrupt number of the MU used to indicate data availability.
uint8_t	mu_id	index of the MU as per PLAT point of view.
uint8_t	tz	Trust Zone (secure) flag - Reflects the XRDC configuration on the MU.
uint8_t	did	DID associated with the MU in the XRDC configuration.

6.9.3 Typedef Documentation

6.9.3.1 she_hdl_t

```
typedef uint32_t she_hdl_t
```

Define the SHE handle type

6.9.4 Function Documentation

6.9.4.1 she_session_hdl_to_ptr()

```
struct she_session_hdl_s* she_session_hdl_to_ptr (
            uint32_t hdl )
```

Returns pointer to the session handle

Parameters

hdl	identifying the session handle.

Returns

pointer to the session handle.

6.9.4.2 delete_she_session()

```
void delete_she_session (
            struct she_session_hdl_s * s_ptr )
```

Delete the session

Parameters

s_ptr	pointer identifying the session.

6.9.4.3 add_she_session()

```
struct she_session_hdl_s* add_she_session (
            void  )
```

Add the session

Returns

pointer to the session.

6.9.4.4 she_service_hdl_to_ptr()

```
struct she_service_hdl_s* she_service_hdl_to_ptr (
            uint32_t hdl )
```

Returns pointer to the service handle

Parameters

hdl	identifying the session handle.

Returns

pointer to the service handle.

6.9.4.5 delete_she_service()

```
void delete_she_service (
            struct she_service_hdl_s * s_ptr )
```

Delete the service

Parameters

s_ptr	pointer identifying the service.

6.9.4.6 add_she_service()

```
struct she_service_hdl_s* add_she_service (
            struct she_session_hdl_s * session )
```

Add the service

Returns

pointer to the service.

6.9.4.7 she_open_session()

```
she_err_t she_open_session (
            open_session_args_t * args,
            she_hdl_t * session_hdl )
```

Parameters

args	pointer to the structure containing the function arguments.
session_hdl	pointer to where the session handle must be written.

Returns

error code.

6.9.4.8 she_close_session()

```
she_err_t she_close_session (
            she_hdl_t session_hdl )
```

Terminate a previously opened session. All the services opened under this session are closed as well

Parameters

session_hdl	pointer to the handle identifying the session to be closed.

Returns

error code.

6.10 SHE keys

Identifiers for SHE keys.

Macros

- #define **SHE_MASTER_ECU_KEY** (0x01)
- #define **SHE_BOOT_MAC_KEY** (0x02)
- #define **SHE_BOOT_MAC** (0x03)
- #define **SHE_KEY_1** (0x04)
- #define **SHE_KEY_2** (0x05)
- #define **SHE_KEY_3** (0x06)
- #define **SHE_KEY_4** (0x07)
- #define **SHE_KEY_5** (0x08)
- #define **SHE_KEY_6** (0x09)
- #define **SHE_KEY_7** (0x0a)
- #define **SHE_KEY_8** (0x0b)
- #define **SHE_KEY_9** (0x0c)
- #define **SHE_KEY_10** (0x0d)
- #define **SHE_RAM_KEY** (0x0e)
- #define SHE_KEY_SIZE_IN_BYTES 16u

  SHE keys are 128 bits (16 bytes) long.
- #define M1_M3_M5_KEY_SIZE_IN_WORDS (SHE_KEY_SIZE_IN_BYTES >> 2)

  size of M1, M3 ad M5 in words.
- #define M2_M4_KEY_SIZE_IN_WORDS (SHE_KEY_SIZE_IN_BYTES >> 1)

  size of M2 ad M4 in words.

6.10.1 Detailed Description

Identifiers for SHE keys.

Refer SHE specification for more information.

6.11 SHE+ key extension

Identifiers for the SHE key extension.

Macros

- #define SHE_KEY_DEFAULT (0x00)

  no key extension: keys from 0 to 10 as defined in SHE specification.
- #define SHE_KEY_N_EXT_1 (0x10)

  keys 11 to 20.
- #define SHE_KEY_N_EXT_2 (0x20)

  keys 21 to 30.
- #define SHE_KEY_N_EXT_3 (0x30)

  keys 31 to 40.
- #define SHE_KEY_N_EXT_4 (0x40)

  keys 41 to 50.

6.11.1 Detailed Description

Identifiers for the SHE key extension.

There are 5 SHE key stores in i.MX95 and 1 SHE keystore in i.MX8DXL
Each key store contains:
1 SECRET KEY (id = 0x0)
1 MASTER ECU KEY (id = 0x1)
1 BOOT MAC KEY (id = 0x2)
1 BOOT MAC (id = 0x3)
10 KEY SLOTS (id = 0x04 to 0xD)
40 extra KEY SLOTS(id = keyID | key_ext,
- keyID from 0x04 to 0xD,
- key_ext like below picture)
1 RAM KEY (id = 0xE)

6.12 Key store

User must open a key store service flow in order to perform all operations on keys.

Data Structures

- struct open_svc_key_store_args_t
- struct op_key_store_reprov_en_args_t

Macros

- #define KEY_STORE_OPEN_FLAGS_DEFAULT 0x0u

  default flags
- #define KEY_STORE_OPEN_FLAGS_CREATE 0x1u

  Create a key store.
- #define KEY_STORE_OPEN_FLAGS_SHE 0x2u

  Target key store is a SHE key store.
- #define KEY_STORE_OPEN_FLAGS_SET_MAC_LEN 0x8u

  Check min mac length.
- #define KEY_STORE_OPEN_FLAGS_SHARED 0x20u

  Target key store is a shared key store (applicable only for i.MX95)
- #define KEY_STORE_OPEN_FLAGS_STRICT_OPERATION 0x80u

  The request is completed only when the key store has been written in the NVM and the monotonic counter has been updated. This flag is applicable for CREATE operation only.
- #define SHE_STORAGE_MIN_MAC_BIT_LENGTH_DEFAULT 32u

  default MAC verification length in bits

Functions

- she_err_t she_open_key_store_service (she_hdl_t session_hdl, open_svc_key_store_args_t *args)
- she_err_t she_close_key_store_service (she_hdl_t key_store_handle)

6.12.1 Detailed Description

User must open a key store service flow in order to perform all operations on keys.

- create a new key store
- perform operations involving keys stored in the key store (ciphering, MAC generation/verification...)
- perform a key store reprovisioning using a signed message. A key store re-provisioning results in erasing all the key stores handled by the SHE.

To grant access to the key store, the caller is authenticated against the domain ID (DID) and Messaging Unit used at the keystore creation, additionally an authentication nonce can be provided.

6.12.2 Data Structure Documentation

6.12.2.1 struct open_svc_key_store_args_t

Structure specifying the open key store service member arguments

Data Fields

uint32_t	key_store_hdl	handle identifying the key store service flow
uint32_t	key_store_identifier	user defined id identifying the key store. Only one key store service can be opened on a given key_store_identifier.
uint32_t	authentication_nonce	user defined nonce used as authentication proof for accessing the key store.
uint8_t	flags	bitmap specifying the services properties.
uint16_t	max_updates_number	maximum number of updates authorized for the key store. • Valid only for create operation. • This parameter has the goal to limit the occupation of the monotonic counter used as anti-rollback protection. • If the maximum number of updates is reached, HSM still allows key store updates but without updating the monotonic counter giving the opportunity for rollback attacks.
uint8_t	min_mac_length	it corresponds to the minimum mac length (in bits) accepted to perform MAC verification operations. Only used upon key store creation when KEY_STORE_FLAGS_SET_MAC_LEN bit is set. It is effective only for MAC verification operations with the mac length expressed in bits. It can be used to replace the default value (32 bits). It impacts all MAC algorithms and all key lengths. It must be different from 0. When in FIPS approved mode values $<$ 32 bits are not allowed. Only used on devices implementing SECO FW.
uint8_t *	signed_message	pointer to signed_message to be sent only in case of key store re-provisioning.
uint16_t	signed_msg_size	size of the signed_message to be sent only in case of key store re-provisioning.

6.12.2.2 struct op_key_store_reprov_en_args_t

Structure describing the key store reprovisioning enable operation arguments

Data Fields

uint8_t *	signed_message	signed content payload
uint32_t	signed_msg_size	signed content payload size in bytes

6.12.3 Function Documentation

6.12.3.1 she_open_key_store_service()

```
she_err_t she_open_key_store_service (
            she_hdl_t session_hdl,
            open_svc_key_store_args_t * args )
```

Open a service flow on the specified key store.

Parameters

session_hdl	SHE handle identifying the current session.
args	pointer to the structure containing the function arguments.

Returns

error code.

6.12.3.2 she_close_key_store_service()

```
she_err_t she_close_key_store_service (
            she_hdl_t key_store_handle )
```

Terminate a previously opened key store service flow

Parameters

key_store_handle	handle identifying the key store service.

Returns

error code.

6.13 CMD_LOAD_KEY

Data Structures

- struct op_key_update_args_t
- struct op_key_update_ext_args_t

Macros

- #define SHE_LOAD_KEY_EXT_FLAGS_STRICT_OPERATION BIT(7)

  User can use this flag to perform multiple updates before writing the key store into the NVM and incrementing the monotonic counter.

Functions

- she_err_t she_key_update (she_hdl_t utils_handle, op_key_update_args_t *args)
- she_err_t she_key_update_ext (she_hdl_t utils_handle, op_key_update_ext_args_t *args)

6.13.1 Detailed Description

Supports two variations of KEY update.

6.13.2 Data Structure Documentation

6.13.2.1 struct op_key_update_args_t

Structure describing the key update operation arguments

Data Fields

uint32_t	utils_handle	Handle to utils service.
uint32_t	key_id	identifier of the key to be used for the operation
uint8_t *	m1	pointer to M1 message
uint8_t	m1_size	size of M1 message - 128 bits
uint8_t *	m2	pointer to M2 message
uint8_t	m2_size	size of M2 message - 256 bits
uint8_t *	m3	pointer to M3 message
uint8_t	m3_size	size of M3 message - 128 bits
uint8_t *	m4	pointer to the output address for M4 message
uint8_t	m4_size	size of M4 message - 256 bits
uint8_t *	m5	pointer to the output address for M5 message
uint8_t	m5_size	size of M5 message - 128 bits

6.13.2.2 struct op_key_update_ext_args_t

Structure describing the key update extension operation arguments

Data Fields

uint32_t	utils_handle	Handle to utils service.
uint32_t	key_ext	identifier of the key extension to be used for the operation
uint32_t	key_id	identifier of the key to be used for the operation
uint8_t *	m1	pointer to M1 message
uint8_t	m1_size	size of M1 message - 128 bits
uint8_t *	m2	pointer to M2 message
uint8_t	m2_size	size of M2 message - 256 bits
uint8_t *	m3	pointer to M3 message
uint8_t	m3_size	size of M3 message - 128 bits
uint8_t *	m4	pointer to the output address for M4 message
uint8_t	m4_size	size of M4 message - 256 bits
uint8_t *	m5	pointer to the output address for M5 message
uint8_t	m5_size	size of M5 message - 128 bits
uint8_t	flags	bitmap specifying the operations property

6.13.3 Function Documentation

6.13.3.1 she_key_update()

```
she_err_t she_key_update (
            she_hdl_t utils_handle,
            op_key_update_args_t * args )
```

Update an internal key of SHE based on memory update protocol refer specification for more information. The request is completed only when the new key has been written in the NVM. The monotonic counter is incremented for each successful update.

Parameters

utils_handle	handle identifying the utils service.
args	pointer to the structure containing the function arguments.

Returns

error code

6.13.3.2 she_key_update_ext()

```
she_err_t she_key_update_ext (
            she_hdl_t utils_handle,
            op_key_update_ext_args_t * args )
```

This is an extension of the CMD_LOAD_KEY The functionality of the CMD_LOAD_KEY is extended by adding a flag argument The updates to the key store must be considered as effective only after an operation specifying the flag "STRICT OPERATION" is acknowledged by SHE

The request is completed only when the key store is written in the NVM and the monotonic counter is incremented

Parameters

utils_handle	handle identifying the utils service
args	pointer to the structure containing the function arguments.

Returns

error code

6.14 CMD_LOAD_PLAIN_KEY

Data Structures

- struct op_load_plain_key_args_t

Functions

- she_err_t she_load_plain_key (she_hdl_t utils_handle, op_load_plain_key_args_t *args)

6.14.1 Detailed Description

Key is handed over in plaintext. A plain key can only be loaded into the RAM_KEY slot.

6.14.2 Data Structure Documentation

6.14.2.1 struct op_load_plain_key_args_t

Structure describing the plain key load operation arguments

Data Fields

uint8_t	key[SHE_KEY_SIZE_IN_BYTES]	pointer to plain key

6.14.3 Function Documentation

6.14.3.1 she_load_plain_key()

```
she_err_t she_load_plain_key (
            she_hdl_t utils_handle,
            op_load_plain_key_args_t * args )
```

Load a key as plaintext to the RAM_KEY slot without encryption and verification.

Parameters

utils_handle	pointer to the SHE utils handle
args	pointer to structure contaiing function arguments

Returns

error code

6.15 CMD_GENERATE_CMAC and CMD_VERIFY_CMAC

Data Structures

- struct open_svc_mac_args_t
- struct op_mac_args_t

Macros

- #define SHE_MAC_VERIFICATION_STATUS_SUCCESS ((she_mac_verification_status_t)(0x6C1AA1↩ C6u))
- #define **SHE_OP_MAC_ONE_GO_ALGO_CMAC** ((she_op_mac_one_go_algo_t)(0x09u))
- #define MAC_OP_FLAGS_MAC_VERIFICATION 0u
- #define MAC_OP_FLAGS_MAC_GENERATION BIT(0)
- #define MAC_OP_FLAGS_MAC_LENGTH_IN_BITS BIT(1)

Typedefs

- typedef uint32_t she_mac_verification_status_t
- typedef uint8_t she_op_mac_one_go_algo_t
- typedef op_mac_args_t op_mac_one_go_args_t

Functions

- she_err_t she_open_mac_service (she_hdl_t key_store_hdl, open_svc_mac_args_t ∗args)
- she_err_t she_mac_one_go (she_hdl_t mac_hdl, op_mac_one_go_args_t ∗args, she_mac_verification_status_t ∗status)
- she_err_t she_close_mac_service (she_hdl_t mac_hdl)

6.15.1 Detailed Description

6.15.2 Data Structure Documentation

6.15.2.1 struct open_svc_mac_args_t

Structure describing the mac open service member agruments

Data Fields

uint8_t	flags	bitmap specifying the services properties.
uint32_t	mac_serv_hdl	indicates the mac handle.

6.15.2.2 struct op_mac_args_t

Structure describing the mac operation member agruments

Data Fields

uint32_t	key_identifier	identifier of the key to be used for the operation
uint8_t	algorithm	algorithm to be used for the operation
uint8_t	flags	bitmap specifying the operation attributes
uint8_t *	payload	pointer to the payload area
uint8_t *	mac	pointer to the mac area • MAC generation: Output MAC value • MAC verification: Input MAC value to verify
uint32_t	payload_size	length in bytes of the payload
uint16_t	mac_size	length of the MAC. • Specified in bytes if MAC_OP_FLAGS_MAC_LENGTH_IN_BITS is clear. • Specified in bits when MAC_OP_FLAGS_MAC_LENGTH_IN_BITS is set. Note: – When specified in bytes the mac size cannot be less than 4 bytes. – When specified in bits the mac size cannot be less than: – the key specific min_mac_len setting if specified for this key when generated/injected; or – the min_mac_length value if specified at the key store provisioning. (if a key specific setting was not specified at key generation/injection); or – the default value (32 bit) if a minimum has not been specified using one of the above 2 methods.
uint32_t	verification_status	mac verification status.
uint8_t	svc_flags	bitmap specifying the services properties.

6.15.3 Macro Definition Documentation

6.15.3.1 SHE_MAC_VERIFICATION_STATUS_SUCCESS

```
#define SHE_MAC_VERIFICATION_STATUS_SUCCESS ((she_mac_verification_status_t)(0x6C1AA1C6u))
```

Bit indicating mac verification success status

6.15.3.2 MAC_OP_FLAGS_MAC_VERIFICATION

```
#define MAC_OP_FLAGS_MAC_VERIFICATION 0u
```

Bit indicating mac verify operation

6.15.3.3 MAC_OP_FLAGS_MAC_GENERATION

```
#define MAC_OP_FLAGS_MAC_GENERATION BIT(0)
```

Bit indicating mac generate operation

6.15.3.4 MAC_OP_FLAGS_MAC_LENGTH_IN_BITS

```
#define MAC_OP_FLAGS_MAC_LENGTH_IN_BITS BIT(1)
```

Bit indicating mac generate operation

6.15.4 Typedef Documentation

6.15.4.1 she_mac_verification_status_t

```
typedef uint32_t she_mac_verification_status_t
```

Bitmap describing the mac verification status

6.15.4.2 she_op_mac_one_go_algo_t

```
typedef uint8_t she_op_mac_one_go_algo_t
```

Bitmap describing the mac one go operation permitted algorithm

6.15.4.3 op_mac_one_go_args_t

```
typedef op_mac_args_t op_mac_one_go_args_t
```

Structure describing the mac one go operation member agruments

6.15.5 Function Documentation

6.15.5.1 she_open_mac_service()

```
she_err_t she_open_mac_service (
            she_hdl_t key_store_hdl,
            open_svc_mac_args_t * args )
```

Open a mac service flow on SHE V2X

User can call this function only after having opened a key store service flow.

User must open this service in order to perform mac operation

Parameters

key_store_hdl	handle identifying the key store service flow.
args	pointer to the structure containing the function arguments.

Returns

error code

6.15.5.2 she_mac_one_go()

```
she_err_t she_mac_one_go (
            she_hdl_t mac_hdl,
            op_mac_one_go_args_t * args,
            she_mac_verification_status_t * status )
```

Perform mac operation User can call this function only after having opened a mac service flow

For mac verification operations, the verified mac length can be specified in:

- Bits by setting the MAC_OP_FLAGS_MAC_LENGTH_IN_BITS flag,
- if this flag is clear then the mac_length is specified in bytes.

For mac generation operations:

- mac length must be set in bytes, and
- MAC_OP_FLAGS_MAC_LENGTH_IN_BITS flag must be 0

Parameters

mac_hdl	handle identifying the mac service flow.
args	pointer to the structure containing the function arguments.
status	pointer for storing the verification status.

Returns

error code

6.15.5.3 she_close_mac_service()

```
she_err_t she_close_mac_service (
            she_hdl_t mac_hdl )
```

Terminate a previously opened mac service flow

Parameters

mac_hdl	pointer to handle identifying the mac service flow to be closed.

Returns

error code

6.16 CMD_INIT_RNG

Functions

- she_err_t she_open_rng_service (she_hdl_t session_hdl, open_svc_rng_args_t *args)
- she_err_t she_close_rng_service (she_hdl_t rng_handle)

6.16.1 Detailed Description

We need to open RNG service before generating Random numbers after every power cycle/reset.

6.16.2 Function Documentation

6.16.2.1 she_open_rng_service()

```
she_err_t she_open_rng_service (
            she_hdl_t session_hdl,
            open_svc_rng_args_t * args )
```

Initializes the seed and derives a key for the PRNG.

User can call this function only after having opened a session.

Parameters

session_hdl	handle identifying the current SHE session.
args	pointer to the structure containing the function arguments.

Returns

error code

6.16.2.2 she_close_rng_service()

```
she_err_t she_close_rng_service (
            she_hdl_t rng_handle )
```

Terminate a previously opened rng service flow

Parameters

rng_handle	handle identifying the RNG session.

Returns

error code

6.17 CMD_RND

Data Structures

- struct open_svc_rng_args_t
- struct op_get_random_args_t

Macros

- #define SHE_RND_SIZE 16u

  size of random data

Typedefs

- typedef uint8_t **svc_rng_flags_t**

Functions

- she_err_t she_get_random (she_hdl_t rng_handle, op_get_random_args_t *args)

6.17.1 Detailed Description

The random number generator has to be initialized by CMD_INIT_RNG before random numbers can be supplied.

6.17.2 Data Structure Documentation

6.17.2.1 struct open_svc_rng_args_t

Structure detailing session open operation member arguments

Data Fields

svc_rng_flags_t	flags	bitmap indicating the service flow properties
uint8_t	reserved[3]	reserved bits
uint32_t	rng_hdl	rng handle

6.17.2.2 struct op_get_random_args_t

Structure detailing the get random number operation member arguments

Data Fields

uint8_t *	output	pointer to the output area where the random number must be written
uint32_t	random_size	length in bytes of the random number to be provided.

Data Fields

svc_rng_flags_t	svc_flags	bitmap indicating the service flow properties
uint8_t	reserved[3]	reserved bits

6.17.3 Function Documentation

6.17.3.1 she_get_random()

```
she_err_t she_get_random (
            she_hdl_t rng_handle,
            op_get_random_args_t * args )
```

returns a vector of 128 random bits.

Parameters

rng_handle	handle identifying the RNG service
args	pointer to the structure containing the function arguments.

Returns

error code

6.18 CMD_EXTEND_SEED

Data Structures

- struct op_rng_extend_seed_t

Macros

- #define SHE_ENTROPY_SIZE 16u
  size of entropy for SHE

Functions

- she_err_t she_extend_seed (she_hdl_t rng_handle, op_rng_extend_seed_t *args)

6.18.1 Detailed Description

The random number generator has to be initialized by CMD_INIT_RNG before the seed can be extended.

6.18.2 Data Structure Documentation

6.18.2.1 struct op_rng_extend_seed_t

Structure describing the RNG extend seed operation arguments

Data Fields

uint32_t	entropy[4]	entropy to extend seed
uint32_t	entropy_size	entropy size

6.18.3 Function Documentation

6.18.3.1 she_extend_seed()

```
she_err_t she_extend_seed (
            she_hdl_t rng_handle,
            op_rng_extend_seed_t * args )
```

extends the seed of the PRNG by compressing the former seed value and the supplied entropy into a new seed which will be used to generate the following random numbers.

Parameters

rng_handle	handle identifying the RNG service
args	pointer to the structure containing entropy vector (128bits)

Returns

error code

6.19 Shared Buffer

Shared Memory and shared buffer.

Data Structures

- struct op_shared_buf_args_t

6.19.1 Detailed Description

Shared Memory and shared buffer.

This is applicable for SECO only not valid for V2X (i.MX95)

6.19.2 Data Structure Documentation

6.19.2.1 struct op_shared_buf_args_t

Structure describing the get shared buffer operation arguments

Data Fields

uint16_t	shared_buf_offset	offset of the shared buffer in secure memory
uint16_t	shared_buf_size	size in bytes of the allocated shared buffer

6.20 Error codes

Error codes returned by SHE functions.

Macros

- #define SAB_INVALID_MESSAGE 0x01

  Invalid/Unknown message.
- #define SAB_INVALID_ADDRESS 0x02

  Invalid Address.
- #define SAB_UNKNOWN_ID 0x03

  Unknown Id.
- #define SAB_INVALID_PARAM 0x04

  MU sanity check failed / Invalid parameters.
- #define SAB_NVM_ERROR 0x05

  NVM general error.
- #define SAB_OUT_OF_MEMORY 0x06

  Internal memory allocation failed.
- #define SAB_UNKNOWN_HANDLE 0x07

  Unknown handle.
- #define SAB_UNKNOWN_KEY_STORE 0x08

  Key store with provided key store ID does not exist (load operation).
- #define SAB_KEY_STORE_AUTH 0x09

  A key store authentication is failing.
- #define SAB_KEY_STORAGE_ERROR 0x0A

  Key store creation/load failure.
- #define SAB_ID_CONFLICT 0x0B

  A Key store using the same key id already exists (create operation).
- #define SAB_RNG_NOT_STARTED 0x0C

  Internal RNG not started.
- #define SAB_CMD_NOT_SUPPORTED 0x0D

  Functionality not supported on current service configuration.
- #define SAB_INVALID_LIFECYCLE 0x0E

  Invalid lifecycle for requested operation.
- #define SAB_KEY_STORE_CONFLICT 0x0F

  The key store already exists (load operation).
- #define SAB_KEY_STORE_COUNTER 0x10

  Issue occurred while updating the key store counter.
- #define SAB_FEATURE_NOT_SUPPORTED 0x11

  Feature is not supported.
- #define SAB_SELF_TEST_FAILURE 0x12

  Self test execution failed.
- #define SAB_NOT_READY 0x13

  System not ready to accept service request.
- #define SAB_FEATURE_DISABLED 0x14

  Feature disabled.
- #define SAB_FATAL_FAILURE 0xFF

  fatal error

Enumerations

- enum she_err_t {
  SHE_NO_ERROR = 0x00,
  SHE_INVALID_MESSAGE = SAB_INVALID_MESSAGE,
  SHE_INVALID_ADDRESS = SAB_INVALID_ADDRESS,
  SHE_UNKNOWN_ID = SAB_UNKNOWN_ID,
  SHE_INVALID_PARAM = SAB_INVALID_PARAM,
  SHE_NVM_ERRO = SAB_NVM_ERROR,
  SHE_OUT_OF_MEMORY = SAB_OUT_OF_MEMORY,
  SHE_UNKNOWN_HANDLE = SAB_UNKNOWN_HANDLE,
  SHE_UNKNOWN_KEY_STORE = SAB_UNKNOWN_KEY_STORE,
  SHE_KEY_STORE_AUTH = SAB_KEY_STORE_AUTH,
  SHE_KEY_STORAGE_ERROR = SAB_KEY_STORAGE_ERROR,
  SHE_ID_CONFLICT = SAB_ID_CONFLICT,
  SHE_RNG_NOT_STARTED = SAB_RNG_NOT_STARTED,
  SHE_CMD_NOT_SUPPORTED = SAB_CMD_NOT_SUPPORTED,
  SHE_INVALID_LIFECYCLE = SAB_INVALID_LIFECYCLE,
  SHE_KEY_STORE_CONFLICT = SAB_KEY_STORE_CONFLICT,
  SHE_KEY_STORE_COUNTER = SAB_KEY_STORE_COUNTER,
  SHE_FEATURE_NOT_SUPPORTED = SAB_FEATURE_NOT_SUPPORTED,
  SHE_SELF_TEST_FAILURE = SAB_SELF_TEST_FAILURE,
  SHE_NOT_READY = SAB_NOT_READY,
  SHE_FEATURE_DISABLED = SAB_FEATURE_DISABLED,
  SHE_UNKNOWN_WARNING = 0x27,
  SHE_SEQUENCE_ERROR_RATING = 0xD1,
  SHE_KEY_NOT_AVAILABLE_RATING = 0xD2,
  SHE_KEY_INVALID_RATING = 0xD3,
  SHE_KEY_EMPTY_RATING = 0xD4,
  SHE_NO_SECURE_BOOT_RATING = 0xD5,
  SHE_KEY_WRITE_PROTECTED_RATING = 0xD6,
  SHE_KEY_UPDATE_ERROR_RATING = 0xD7,
  SHE_RNG_SEED_RATING = 0xD8,
  SHE_NO_DEBUGGING_RATING = 0xD9,
  SHE_BUSY_RATING = 0xDA,
  SHE_MEMORY_FAILURE_RATING = 0xDB,
  SHE_GENERAL_ERROR = 0xDC,
  SHE_LIB_ERROR = 0xEF,
  SHE_FATAL_FAILURE = SAB_FATAL_FAILURE }

6.20.1 Detailed Description

Error codes returned by SHE functions.

Details on all error codes returned by SHE APIs

6.20.2 Enumeration Type Documentation

6.20.2.1 she_err_t

`enum she_err_t`

Error codes returned by SHE functions.

Enumerator

SHE_NO_ERROR	Success.
SHE_INVALID_MESSAGE	Invalid/Unknown message.
SHE_INVALID_ADDRESS	Invalid Address.
SHE_UNKNOWN_ID	Unknown Id.
SHE_INVALID_PARAM	MU sanity check failed / Invalid parameters.
SHE_NVM_ERRO	NVM general error.
SHE_OUT_OF_MEMORY	Internal memory allocation failed.
SHE_UNKNOWN_HANDLE	Unknown handle.
SHE_UNKNOWN_KEY_STORE	Key store with provided key store ID does not exist (load operation).
SHE_KEY_STORE_AUTH	A key store authentication is failing.
SHE_KEY_STORAGE_ERROR	Key store creation/load failure.
SHE_ID_CONFLICT	A Key store using the same key id already exists (create operation).
SHE_RNG_NOT_STARTED	Internal RNG not started.
SHE_CMD_NOT_SUPPORTED	Functionality not supported on current service configuration.
SHE_INVALID_LIFECYCLE	Invalid lifecycle for requested operation.
SHE_KEY_STORE_CONFLICT	The key store already exists (load operation).
SHE_KEY_STORE_COUNTER	Issue occurred while updating the key store counter.
SHE_FEATURE_NOT_SUPPORTED	Feature is not supported.
SHE_SELF_TEST_FAILURE	Self test execution failed.
SHE_NOT_READY	System not ready to accept service request.
SHE_FEATURE_DISABLED	Feature disabled.
SHE_UNKNOWN_WARNING	SHE Unknown Warning.
SHE_SEQUENCE_ERROR_RATING	Invalid sequence of commands.
SHE_KEY_NOT_AVAILABLE_RATING	Key is locked.
SHE_KEY_INVALID_RATING	Key not allowed for the given operation.
SHE_KEY_EMPTY_RATING	Key has not been initialized yet.
SHE_NO_SECURE_BOOT_RATING	Conditions for a secure boot process are not met.
SHE_KEY_WRITE_PROTECTED_RATING	Memory slot for this key has been write-protected.
SHE_KEY_UPDATE_ERROR_RATING	Key update did not succeed, errors in verification of message.
SHE_RNG_SEED_RATING	The seed has not been initialized.
SHE_NO_DEBUGGING_RATING	Internal debugging is not possible.
SHE_BUSY_RATING	SHE is busy.
SHE_MEMORY_FAILURE_RATING	Memory Error.
SHE_GENERAL_ERROR	SHE General error.
SHE_LIB_ERROR	SHE library error.
SHE_FATAL_FAILURE	fatal error

6.21 Utils

The SHE utils service flow can be opened only after opening SHE key storage handle.

Data Structures

- struct op_open_utils_args_t

Functions

- she_err_t she_open_utils (she_hdl_t key_store_handle, op_open_utils_args_t *args)
- she_err_t she_close_utils (she_hdl_t utils_handle)

6.21.1 Detailed Description

The SHE utils service flow can be opened only after opening SHE key storage handle.

User must open a SHE utils service flow in order to perform all operations:

- Create a utils handle
- perform SHE key update extension
- update SHE plain key
- export SHE plain key
- get SHE identity (UID)
- get SHE status register
- perform MAC generation and verification in fast mode for a SHE session

6.21.2 Data Structure Documentation

6.21.2.1 struct op_open_utils_args_t

Structure describing the open utils service operation arguments

Data Fields

uint32_t	utils_handle	

6.21.3 Function Documentation

6.21.3.1 she_open_utils()

```
she_err_t she_open_utils (
            she_hdl_t key_store_handle,
            op_open_utils_args_t * args )
```

Open SHE utils service flow on the specified key store. The SHE utils service flow can be opened only after opening SHE key storage handle.

Parameters

key_store_handle	handle identifying the key store service.
args	pointer to the structure containing the function arguments.

Returns

error code.

6.21.3.2 she_close_utils()

```
she_err_t she_close_utils (
            she_hdl_t utils_handle )
```

Terminate a previously opened utils service flow

Parameters

utils_handle	handle identifying the utils service.

Returns

error code.

6.22 last rating code

Functions

- uint32_t she_get_last_rating_code (she_hdl_t session_hdl)

6.22.1 Detailed Description

We can get last rating code (error) using the below described function. This information is helpful during debugging an issue

6.22.2 Function Documentation

6.22.2.1 she_get_last_rating_code()

```
uint32_t she_get_last_rating_code (
            she_hdl_t session_hdl )
```

Report rating code from last command

SHE API defines standard errors that should be returned by API calls. Error code reported by SECO/V2X are "translated" to these SHE error codes. This API allow user to get the error code reported by SECO/V2X for the last command before its translation to SHE error codes. This should be used for debug purpose only.

Parameters

session_hdl	SHE session handle

Returns

rating code reported by last command

6.23 CMD_CANCEL

Functions

- void she_cmd_cancel (void)

6.23.1 Detailed Description

To cancel any running command, we can use below described command. This is helpful if we have mistakenly send an incorrect command and want to stop the execution of that command or discard the results if command has already been executed.

6.23.2 Function Documentation

6.23.2.1 she_cmd_cancel()

```
void she_cmd_cancel (
             void  )
```

interrupt any given function and discard all calculations and results.

6.24 Global Info

Modules

- IMX8DXL
- IMX95

Macros

- #define **SOC_IMX8QXP** 0x2
- #define **SOC_IMX8DXL** 0xe
- #define **IMX8DXL_DL1** 0x1
- #define **IMX8DXL_DL2** 0x2
- #define **IMX8DXL_DL3** 0x4
- #define **SOC_IMX8ULP** 0x84d
- #define **SOC_IMX91** 0x9100
- #define **SOC_IMX93** 0x9300
- #define **SOC_IMX95** 0x9500
- #define **SOC_REV_A0** 0xa000
- #define **SOC_REV_A1** 0xa100
- #define **SOC_REV_A2** 0xa200
- #define **SOC_REV_B0** 0xb000
- #define **SOC_REV_C0** 0xc000
- #define **GINFO_LIB_VERSION_LEN** 16
- #define **GINFO_NVM_VERSION_LEN** 16
- #define **GINFO_COMMIT_ID_SZ** 40
- #define **HSM_API_VERSION_1** 0x1
- #define **HSM_API_VERSION_2** 0x2
- #define SOC_LF_FAB_DEFAULT 0x1

  Default fab mode chip lifecycle.
- #define SOC_LF_FAB_MODE 0x2

  Fab mode chip lifecycle.
- #define SOC_LF_NO_NXP_SECRETS 0x4

  No NXP secrets chip lifecycle.
- #define SOC_LF_WITH_NXP_SECRETS 0x8

  With NXP secrets chip lifecycle.
- #define SOC_LF_SCU_FW_CLOSED 0x10

  SCU FW Closed chip lifecycle.
- #define SOC_LF_SECO_FW_CLOSED 0x20

  SECO FW Closed chip lifecycle.
- #define SOC_LF_CLOSED 0x40

  Closed chip lifecycle.
- #define SOC_LF_CLOSED_WITH_NXP_FW 0x80

  Closed with NXP FW chip lifecycle.
- #define SOC_LF_PARTIAL_FIELD_RET 0x100

  Partial Field Return chip lifecycle.
- #define SOC_LF_FIELD_RET 0x200

  Field Return chip lifecycle.
- #define SOC_LF_NO_RET 0x400

  No Return chip lifecycle.
- #define SOC_LF_BLANK 0x1

Blank chip lifecycle.

- #define SOC_LF_DEFAULT_FAB 0x2

  Fab Default chip lifecycle.
- #define SOC_LF_FAB 0x4

  Fab chip lifecycle.
- #define SOC_LF_NXP_PROVISIONED 0x8

  NXP Provisioned chip lifecycle.
- #define SOC_LF_OEM_OPEN 0x10

  OEM Open chip lifecycle.
- #define SOC_LF_OEM_SECURE_WORLD_CLOSED 0x20

  OEM Secure-World-Closed chip lifecycle.
- #define SOC_LF_OEM_CLOSED 0x40

  OEM Closed chip lifecycle.
- #define SOC_LF_OEM_LOCKED 0x80

  OEM Locked chip lifecycle.
- #define SOC_LF_FIELD_RET_OEM 0x100

  Field Return OEM chip lifecycle.
- #define SOC_LF_FIELD_RET_NXP 0x200

  Field Return NXP chip lifecycle.
- #define SOC_LF_BRICKED 0x400

  Bricked chip lifecycle.

Functions

- void populate_global_info (uint32_t session_hdl)
- void show_global_info (void)
- bool is_global_info_populated (void)
- uint8_t se_get_board_type (void)
- uint16_t se_get_soc_id (void)
- uint16_t se_get_soc_rev (void)
- uint16_t se_get_chip_lifecycle (void)
- uint8_t se_get_fips_mode (void)
- uint8_t se_get_lib_newness_ver (void)
- uint8_t se_get_lib_major_ver (void)
- uint8_t se_get_lib_minor_ver (void)
- uint8_t se_get_nvm_newness_ver (void)
- uint8_t se_get_nvm_major_ver (void)
- uint8_t se_get_nvm_minor_ver (void)
- const char * se_get_commit_id (void)
- const char * se_get_lib_version (void)
- const char * se_get_nvm_version (void)
- const char * get_board_type_str (uint8_t board_type)
- const char * get_soc_id_str (uint16_t soc_id)
- const char * get_soc_rev_str (uint16_t soc_rev)
- const char * get_soc_lf_str (uint16_t lifecycle)
- void se_get_info (uint32_t session_hdl, op_get_info_args_t *args)
- void se_get_soc_info (uint32_t session_hdl, uint16_t *soc_id, uint16_t *soc_rev, uint8_t *board_type)
- uint32_t get_global_info_flags (void)
- void set_global_info_flags (uint32_t flags)

6.24.1 Detailed Description

6.24.2 Function Documentation

6.24.2.1 populate_global_info()

```
void populate_global_info (
            uint32_t session_hdl )
```

Populate the Global Info structure

Parameters

session_hdl	identifying the session.

6.24.2.2 show_global_info()

```
void show_global_info (
            void  )
```

Print the Global Info of library

6.24.2.3 is_global_info_populated()

```
bool is_global_info_populated (
            void  )
```

Get the status of Global Info, if populated or not.

6.24.2.4 se_get_board_type()

```
uint8_t se_get_board_type (
            void  )
```

Get board type.

6.24.2.5 se_get_soc_id()

```
uint16_t se_get_soc_id (
            void  )
```

Get SoC ID.

6.24.2.6 se_get_soc_rev()

```
uint16_t se_get_soc_rev (
            void  )
```

Get SoC Revision.

6.24.2.7 se_get_chip_lifecycle()

```
uint16_t se_get_chip_lifecycle (
            void  )
```

Get Chip-lifecycle.

6.24.2.8 se_get_fips_mode()

```
uint8_t se_get_fips_mode (
            void  )
```

Get Fips mode.

6.24.2.9 se_get_lib_newness_ver()

```
uint8_t se_get_lib_newness_ver (
            void  )
```

Get library newness version.

6.24.2.10 se_get_lib_major_ver()

```
uint8_t se_get_lib_major_ver (
            void  )
```

Get library major version.

6.24.2.11 se_get_lib_minor_ver()

```
uint8_t se_get_lib_minor_ver (
            void  )
```

Get library minor version.

6.24.2.12 se_get_nvm_newness_ver()

```
uint8_t se_get_nvm_newness_ver (
            void  )
```

Get NVM newness version.

6.24.2.13 se_get_nvm_major_ver()

```
uint8_t se_get_nvm_major_ver (
            void  )
```

Get NVM major version.

6.24.2.14 se_get_nvm_minor_ver()

```
uint8_t se_get_nvm_minor_ver (
            void  )
```

Get NVM minor version.

6.24.2.15 se_get_commit_id()

```
const char* se_get_commit_id (
            void  )
```

Get Build commit id.

6.24.2.16 se_get_lib_version()

```
const char* se_get_lib_version (
            void  )
```

Get library version string.

6.24.2.17 se_get_nvm_version()

```
const char* se_get_nvm_version (
            void  )
```

Get NVM version string.

6.24.2.18 get_board_type_str()

```
const char* get_board_type_str (
            uint8_t board_type )
```

Get the string representating board type

Parameters

board_type	Board Type fetched from Global Info

Returns

String represention of the board type

6.24.2.19 get_soc_id_str()

```
const char* get_soc_id_str (
            uint16_t soc_id )
```

Get the string representating SoC ID

Parameters

soc_id	SoC ID fetched from Global Info

Returns

String represention of the SoC ID

6.24.2.20 get_soc_rev_str()

```
const char* get_soc_rev_str (
            uint16_t soc_rev )
```

Get the string representating SoC Revision

Parameters

soc_rev	SoC Revision fetched from Global Info

Returns

String represention of the SoC Revision

6.24.2.21 get_soc_lf_str()

```
const char* get_soc_lf_str (
            uint16_t lifecycle )
```

Get the string representation of the Chip Lifecycle

Parameters

lifecycle	value fetched from Global Info, which will vary across SOCs

Returns

a string represention of Lifecycle

6.24.2.22 se_get_info()

```
void se_get_info (
            uint32_t session_hdl,
            op_get_info_args_t * args )
```

Get Info for Global Info setup

6.24.2.23 se_get_soc_info()

```
void se_get_soc_info (
            uint32_t session_hdl,
            uint16_t * soc_id,
            uint16_t * soc_rev,
            uint8_t * board_type )
```

Get SoC Info for Global Info setup

6.24.2.24 get_global_info_flags()

```
uint32_t get_global_info_flags (
            void  )
```

Get Global Info flags

6.24.2.25 set_global_info_flags()

```
void set_global_info_flags (
            uint32_t flags )
```

Set Global Info flags

6.25 IMX8DXL

Global Info

i.MX8DXL Lifecycle

- SOC_LF_FAB_DEFAULT
- SOC_LF_FAB_MODE
- SOC_LF_NO_NXP_SECRETS
- SOC_LF_WITH_NXP_SECRETS
- SOC_LF_SCU_FW_CLOSED
- SOC_LF_SECO_FW_CLOSED
- SOC_LF_CLOSED
- SOC_LF_CLOSED_WITH_NXP_FW
- SOC_LF_PARTIAL_FIELD_RET
- SOC_LF_FIELD_RET
- SOC_LF_NO_RET

6.26 IMX95

Global Info

i.MX95 Lifecycle

- SOC_LF_BLANK
- SOC_LF_DEFAULT_FAB
- SOC_LF_FAB
- SOC_LF_NXP_PROVISIONED
- SOC_LF_OEM_OPEN
- SOC_LF_OEM_SECURE_WORLD_CLOSED
- SOC_LF_OEM_CLOSED
- SOC_LF_OEM_LOCKED
- SOC_LF_FIELD_RET_OEM
- SOC_LF_FIELD_RET_NXP
- SOC_LF_BRICKED

Chapter 7

Data Structure Documentation

7.1 time_frame Struct Reference

Data Fields

- struct timespec t_start

  Start timer.
- struct timespec t_end

  End timer.

7.1.1 Detailed Description

Structure detailing the members to store timestamps from kernel

7.1.2 Field Documentation

7.1.2.1 t_start

```
struct timespec time_frame::t_start
```

Start timer.

7.1.2.2 t_end

```
struct timespec time_frame::t_end
```

End timer.

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